

Kayleen Welch

1-413-579-2632

kleen2797@gmail.com

Electrical Engineer

B/T Rochester Institute of Technology

kleen2797.github.io/Portfolio-Site/

Technical skills:

Electronics Hardware Design . Microcontroller integrated software design . Solder certified IPC-A-610 . Schematic Design & Analysis . Breadboard & bench tools: AWG, Oscilloscope, Spectrum Analyzer . Mechanical benchtop tools

Software: C/C++ . Altium Designer . Jira . Agile PM . Arduino . MATLAB . Multisim . Modelsim . Draftsight . Teamcenter NX . MicrosoftOffice Suite & Visio . HTML/CSS

Work Experience

Fisher - Price (Mattel) - Electronics Design Engineer / Buffalo NY, July 2022 - Nov 2025

Led Electronics Design as a member of a cross functional team, handling design and execution of product ideas from concept ideation through manufacturer handoff and production troubleshooting. Responsible for hardware design and fabrication methods, costing and BOM, product software development and execution, and troubleshooting any quality or compliance concerns

IEC Electronics - Design Engineer / Newark NY May 2021 - Feb 2022

Part of a small team, responsible for all aspects of building client's testing equipment. Designed, built, tested and installed several large PCBs and designed and built interfacing cables for client boards, instruments, and in-house testing equipment. Documented designs and system functionality to ISO and ITAR auditing standards, and participated in design reviews for IPC compliance. Additional responsibilities included guiding co-op student work and problem solving for in-house board assembly facilities as required

UR Laser Lab for Energetics - Electrical Engineering Co-op / Rochester NY Jan 2019 - Mar 2020

Worked as a Research & Lab Engineer for the Electrical Engineering Department on projects in various areas from Controls to Measurement tools. Successfully followed projects from design to fabrication, testing and analysis, and implementation

RIT Lab Assistant / Rochester NY Aug 2018 - May 2019

TA for two sections each of DC Circuits and Electronics 1 Labs; assist with equipment issues & technical questions

Relevant Labs

Electronics I & II - Designed, built, simulated, and analyzed advanced networks utilizing semiconductor devices: diodes, BJT, & chip Op-Amp circuits

Microelectronic Design & Fabrication - Designed and Fabricated NPN transistor and resistor network in a Class 1000 Cleanroom Lab

Digital Signal Processing - Optimization of Analog signals by building a digital signal processing filter on a Microprocessor, and documentation in IEEE standard format

DC & AC Circuits - Designed, built, simulated and analyzed various AC and DC circuits to prove fundamental principles of Electronic Engineering

Microcontroller Systems - Wrote programs to operate Arduino robot using Atmega328p Microprocessor and Arduino IDE

Computational Problem Solving - Used C++ to create programs to accomplish a variety of tasks using arrays, vectors, sorts, and C++ library functions

Projects & Hobbies

Women in Technology member Assists with community outreach, participates in meetings, team building exercises and outings. (2018-present)

Hotwheelz Formula Hybrid SAE Electrical team member; designed and built GLV system on all electric Formula One car; placed 3rd at nationals (2015-2016)